CS23E01 / EMBEDDED SYSTEMS AND INTERNET OF THINGS

- Session: July-Nov 2026 (B.E. CSE – VII sem)

Activities for Lab-4:

Topics: I/O operations, logical operations, interrupts – timer based, External interrupts in **Embedded C**.

Software/simulator required: MCU 8051 IDE with SDCC compiler or equivalent.

// For basic Embedded 'C' programs refer to the pdf/material shared.

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1. Demonstrate the working of software delay through LED blinking example.
2. Read a byte from any one of the available ports and write the output to another port.
3. Read two bytes of data from two different ports and add the numbers and save the sum to any register. Display the corresponding SFR's values.
4. Perform simple logic operations AND,OR,XOR on data bytes read from two different ports.
5. Using the C/T bit from TMOD register by changing timer as counter, count the no. of 1's in the input byte at any port and display the result of the count in another register.
6. Demonstrate the working of TIMER interrupts, External hardware interrupts:
    a. Check the time elapsed and update the value of a register.
    b. Assume there is a keyboard input on any port at a random time instance, and run an ISR.